



1M1B – IBM SkillsBuild AI + Sustainability Virtual  
Internship

# ECOSORT AI: A MULTIMODEL ASSISTANT FOR E-WASTE MANAGEMENT

Leveraging AI Vision & RAG for  
Responsible Consumption

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# INTRODUCTION: THE E-WASTE CHALLENGE IN INDIA

India generates over 3.2 million tonnes of e-waste annually, yet less than 5% is formally recycled. Toxic heavy metals such as Lead, Mercury, and Cadmium pose severe health and environmental risks. EcoSort AI leverages Multimodal AI Vision and RAG to empower individuals with instant, responsible e-waste disposal decisions.

✓ 3.2M+ Tonnes of E-Waste Generated Annually in India

✓ Less Than 5% of E-Waste is Formally Recycled

✓ Toxic Metals: Lead, Mercury & Cadmium Present





# PROBLEM STATEMENT

E-waste mismanagement poses serious environmental and health risks across India. Millions of households and institutions lack the knowledge, tools, and access needed to dispose of electronic waste responsibly — leading to toxic contamination and inefficient recycling outcomes.



Lack of Awareness: Hazardous Items Go Unrecognized



Lack of Accessibility to Nearest Collection Centers



Inefficient Segregation at Landfills & Waste Sites





# OBJECTIVES

EcoSort AI is designed with clear, impact-driven goals to tackle the growing e-waste crisis in India through cutting-edge AI technology, responsible practices, and user-centric design aligned with global sustainability standards.



Classify E-Waste via AI Vision (batteries, chargers, CFL bulbs, PCBs)



Provide RAG-Powered Disposal Guidance & Locate Nearest Recyclers



Educate on Hazards & Promote UN SDG 12 Responsible Consumption

# DATASET OVERVIEW

Custom E-Waste Image Dataset with 5,100 labeled images across 6 categories. Sources include Kaggle, Google Open Images, and manual collection. RAG knowledge base includes municipal policies, recycler contacts, and safety guidelines.

01

Lithium-ion Batteries: 1,200 | Alkaline Batteries: 800 | Mobile Chargers: 1,000

02

CFL Bulbs: 900 | PCBs (Circuit Boards): 700 | Miscellaneous: 500

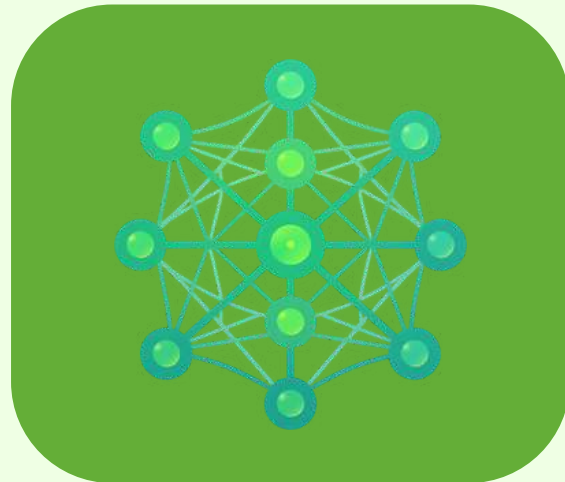
03

Total: 5,100 Images — Kaggle, Google Open Images & Manual Collection





# TECHNOLOGY STACK: AI VISION



**Deep Learning  
Frameworks**



**Vision Model  
Architectures**



**Image Processing &  
Training**

EcoSort AI leverages TensorFlow and PyTorch as its core deep learning frameworks for building and training e-waste classification models. The vision backbone uses ResNet-50 with transfer learning for high accuracy across 6 e-waste categories, while MobileNetV2 provides a lightweight alternative optimized for faster inference. Models are trained with the Adam optimizer over 25 epochs, achieving 92.4% classification accuracy on the custom 5,100-image dataset.

# METHODOLOGY

A Design Thinking approach was used to develop EcoSort AI, ensuring the solution is user-centric, practical, and impactful. Each phase informed the next, from understanding real user pain points to building and refining a working prototype.



01

Empathize: 90% store electronics at home without guidance

02

Define & Ideate: Combine Vision AI + RAG for real-time advice

03

Prototype & Test: Refined with mentor feedback & iteration

# DATA PROCESSING

Robust preprocessing pipelines prepare both image and text data for accurate e-waste classification and retrieval-augmented generation.

Resize to 224x224px, Normalize & Augment Images

Label Encoding & Document Chunking (500 words)

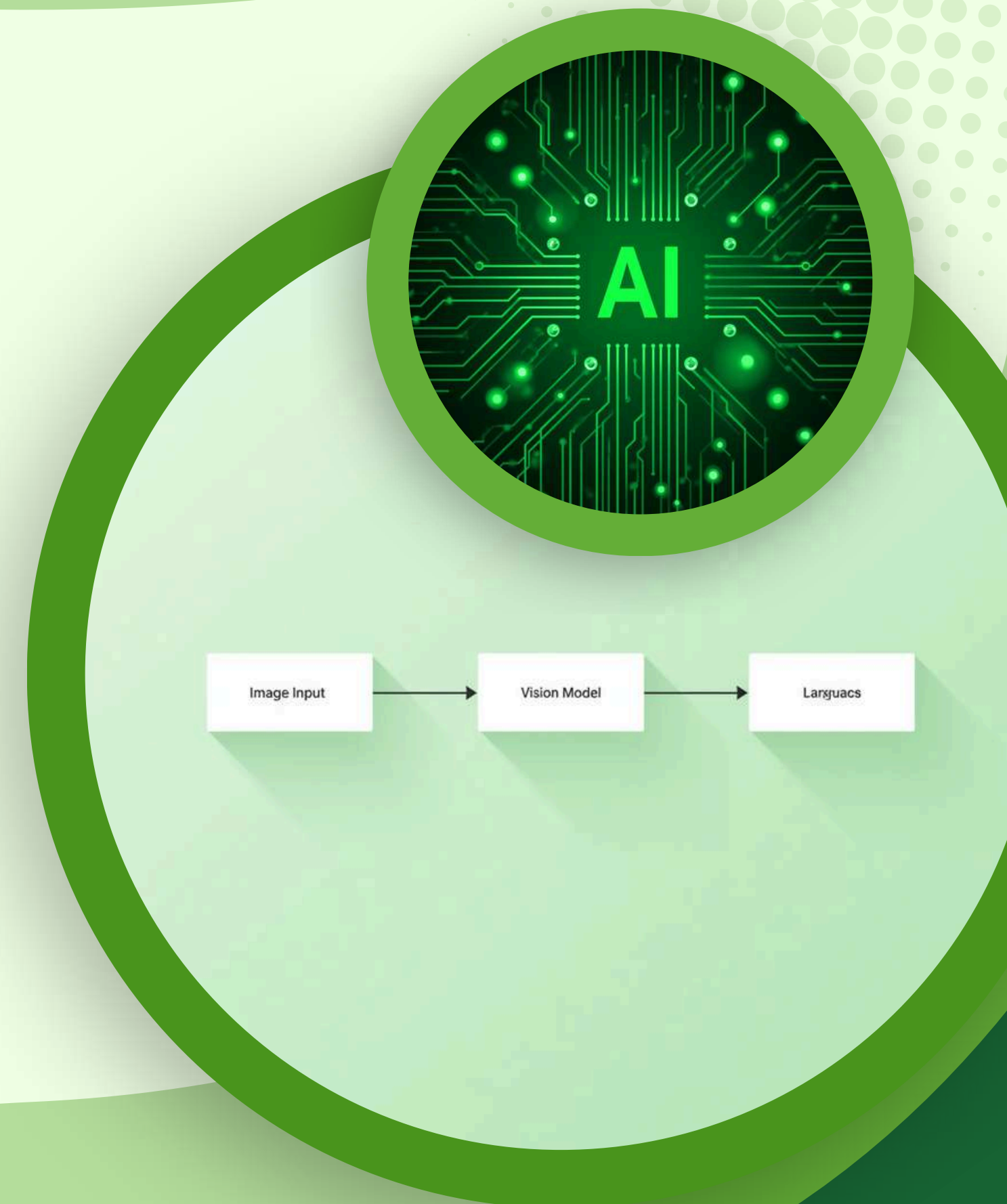
Embeddings Generated & Stored in FAISS Vector DB





# SYSTEM ARCHITECTURE

EcoSort AI follows a seamless multimodal pipeline: the user uploads an e-waste image with an optional question, triggering a chain of AI-powered modules that deliver accurate classification, policy retrieval, and actionable disposal guidance in real time.



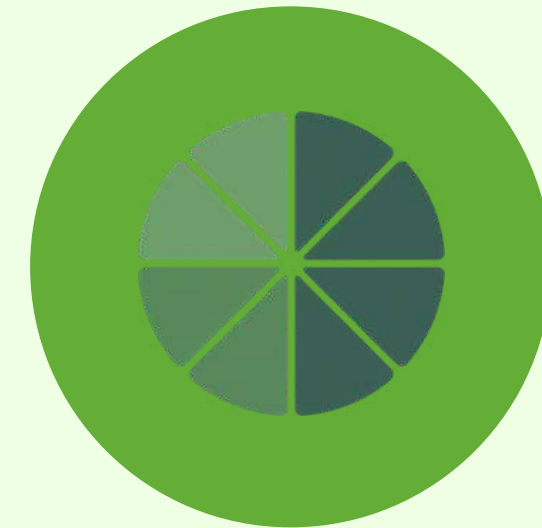
# MACHINE LEARNING MODEL DETAILS



**Base Model &  
Architecture**



**Training  
Configuration**



**Output &  
Classification**

Base Model: ResNet-50 with Transfer Learning — fine-tuned to classify 6 e-waste categories (Lithium-ion Batteries, Alkaline Batteries, Mobile Chargers, CFL Bulbs, PCBs, Miscellaneous). Output layer uses Softmax probabilities for multi-class classification. Training: 25 Epochs | Batch Size: 32 | Learning Rate: 0.001 | Optimizer: Adam | Loss: Categorical Cross-Entropy | Dataset Split: 80% Train / 20% Validation. Early stopping applied to prevent overfitting.



# MODEL PERFORMANCE METRICS



Accuracy: 92.4% — High Classification Rate



Precision: 91.8% & Recall: 90.5%



F1-Score: 91.1% — Robust & Balanced Model



# SAMPLE RESULTS

EcoSort AI correctly identifies hazardous e-waste items and delivers actionable disposal guidance. When a CFL bulb image was uploaded, the AI detected Mercury hazard, advised sealed bag disposal, and flagged the item as high-risk. For a laptop battery query, the system warned of explosion risk and located GreenHub Recyclers just 2.3 km away — demonstrating real-time, location-aware e-waste assistance.



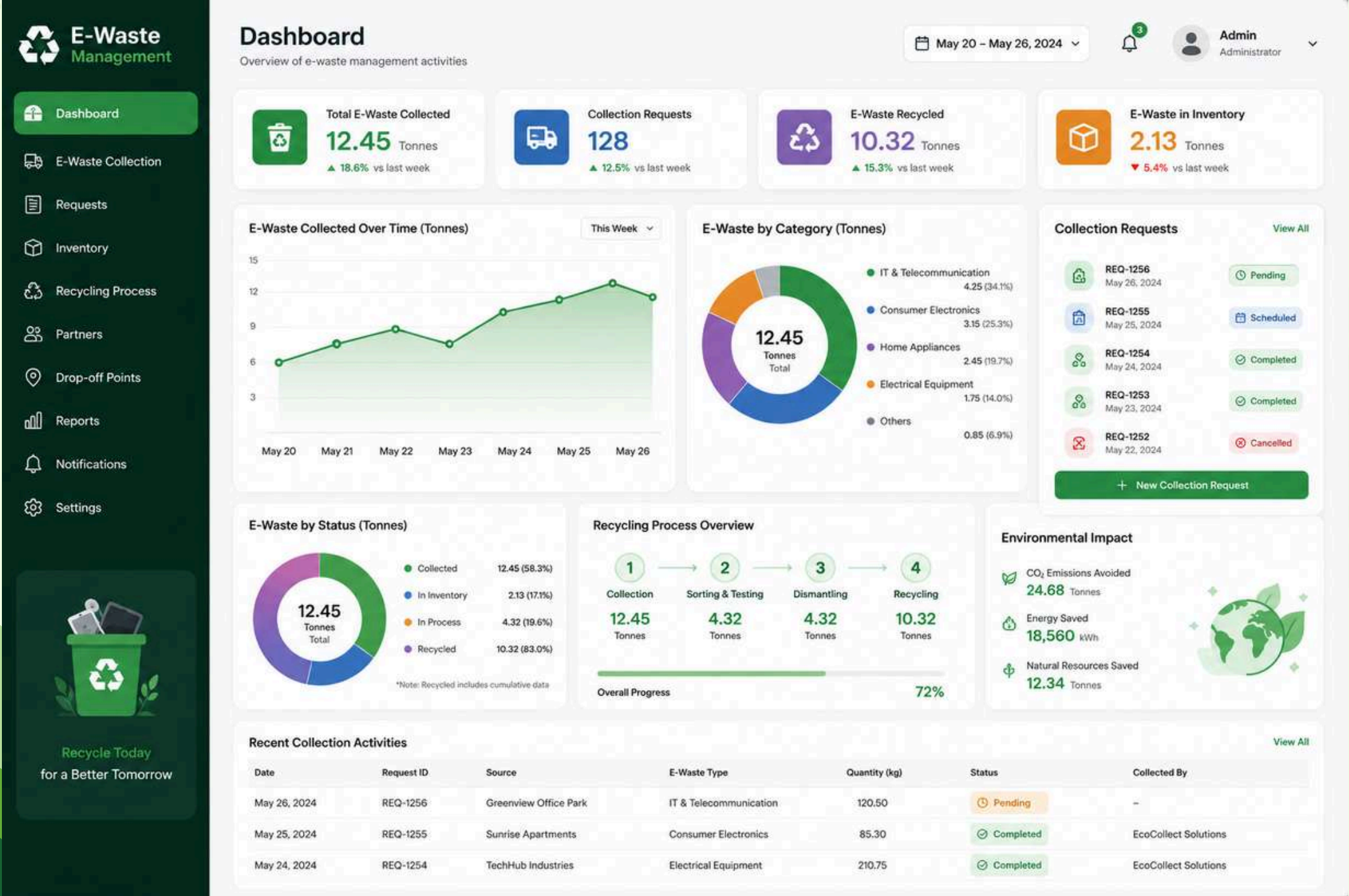
CFL Bulb: Mercury  
Hazard Detected

Sealed Bag  
Disposal Advised

Laptop Battery:  
Recycler Located



# UI DASHBOARD OVERVIEW





# REAL-WORLD APPLICATIONS

EcoSort AI is designed for scalable deployment across multiple real-world sectors, enabling smarter e-waste management and promoting responsible consumption in communities, institutions, and government initiatives.

✓ Municipal Waste Centers & Smart City Dashboards

✓ College Campuses & Recycling Companies

✓ Government Awareness & Sustainability Campaigns





# FUTURE SCOPE

EcoSort AI is designed to evolve with emerging technologies. The following enhancements will expand its capabilities, reach, and impact — transforming it into a comprehensive, scalable e-waste management platform for individuals, municipalities, and smart cities alike.



YOLO Detection & GPS-Tracked Collection Vans for Real-Time Multi-Item Identification & Dynamic Pickups



Blockchain Traceability & Gamification Rewards for Transparent Recycling & User Engagement



Multilingual Support (Hindi, Tamil, Kannada) & Raspberry Pi Edge Deployment for Offline Use

# CONCLUSION

EcoSort AI demonstrates the transformative power of AI as a sustainability tool, combining Multimodal Vision and Retrieval-Augmented Generation (RAG) to deliver an end-to-end e-waste management solution.

- AI + RAG delivers real-time classification and disposal guidance
- Key learning: AI requires logical workflow, responsible practices, and user-centric design
- Small projects can create measurable environmental impact at scale
- Bridges the gap between awareness and actionable e-waste decisions
- Aligns with UN SDG 12: Responsible Consumption and Production
- Supports IBM SkillsBuild AI + Sustainability Internship goals

EcoSort AI is a step toward a cleaner, greener, and more responsible digital future.







# THANK YOU FOR YOUR ATTENTION!

